

Chapter 1

RTL Design & Implementation of Digital Tachometer

1.1 IP Block 4: Digital Tachometer (Frequency-to-Digital Converter)

In mixed-signal control systems, reading the rotational speed of a physical motor requires translating the asynchronous analog pulses of a physical sensor (such as a Hall-effect or optical encoder) into a precise digital format.

To bridge this gap, a **Digital Tachometer** was developed as the fifth IP block. Instead of utilizing a slow "Gate Time" frequency measurement approach, this module utilizes a high-speed "Period Measurement" architecture, allowing it to calculate instantaneous motor speeds in a fraction of a millisecond without bottlenecking the system.

1.1.1 Architectural Theory & Hardware Advantages

The Period Measurement architecture acts as a highly precise digital stopwatch. It calculates speed by measuring the exact time elapsed between two consecutive motor sensor pulses using a high-frequency system clock (`clk_sys`).

- **Asynchronous Synchronization:** Because the physical motor rotates independently of the FPGA's system clock, raw sensor pulses are inherently asynchronous. Feeding them directly into a synchronous counter causes severe metastability. To mitigate this, a 3-stage flip-flop synchronizer is implemented on the front end. This aligns the external pulse to the internal clock domain and guarantees a clean, stable rising-edge detection pulse.
- **Continuous Tracking:** The hardware relies on a 16-bit free-running counter. The exact microsecond a synchronized motor pulse arrives, the hardware simultaneously captures the current count, flags a `data_valid` interrupt, and resets the counter to zero to immediately begin measuring the next rotation.

1.1.2 Real-Time Applications

This microsecond-latency period measurement logic is foundational in high-performance physical control loops:

- **Electric Vehicles (EVs) & ABS:** Instantly detects wheel slip by measuring the elongated period between Hall-effect pulses, allowing traction control systems to cut power before a spin-out occurs.

- **Industrial CNC & Robotics:** Provides zero-latency speed feedback to PID controllers, allowing heavy-duty servo motors to instantaneously compensate for sudden physical loads.
- **UAV Flight Controllers:** Verifies physical BLDC motor RPMs against commanded speeds thousands of times per second to stabilize drones against wind shear.
- **Low-RPM Generation (Wind Turbines):** Accurately measures highly massive, slow-moving generators where traditional "pulses-per-second" gate counters would fail to register a reading.

1.1.3 Complete Synthesizable Verilog RTL Description

The following synthesizable RTL implements the anti-metastability synchronizer and the high-speed measurement counter.

```
'timescale 1ns / 1ps

module digital_tachometer (
    input wire      clk_sys,      // High-speed system clock (e.g., 10 MHz)
    input wire      rst_n,        // Active-low asynchronous reset
    input wire      sensor_in,    // Asynchronous pulse from the motor/sensor

    output reg [15:0] period_count, // Output: Number of sys_clks between sensor pulses
    output reg      data_valid      // Output: Pulses HIGH for 1 tick when reading is ready
);

//-----
// 1. ASYNCHRONOUS SENSOR SYNCHRONIZER & EDGE DETECTOR
//-----
reg sync_1, sync_2, sync_3;

always @(posedge clk_sys or negedge rst_n) begin
    if (!rst_n) begin
        sync_1 <= 1'b0;
        sync_2 <= 1'b0;
        sync_3 <= 1'b0;
    end else begin
        sync_1 <= sensor_in;
        sync_2 <= sync_1;
        sync_3 <= sync_2;
    end
end

// A rising edge is detected when current state is 1, but previous was 0
wire sensor_rising_edge = (sync_2 == 1'b1) && (sync_3 == 1'b0);

//-----
// 2. PERIOD MEASUREMENT COUNTER
//-----
reg [15:0] counter;

always @(posedge clk_sys or negedge rst_n) begin
    if (!rst_n) begin
        counter      <= 16'd0;
        period_count <= 16'd0;
        data_valid    <= 1'b0;
    end else begin
```

```

    if (sensor_rising_edge) begin
        // Rotation complete: Capture time, output it, and reset counter
        period_count <= counter;
        counter      <= 16'd0;
        data_valid   <= 1'b1;
    end else begin
        // Still spinning: Keep counting system clock ticks
        data_valid <= 1'b0;

        if (counter != 16'hFFFF) begin // Prevent integer overflow
            counter <= counter + 16'd1;
        end
    end
end
end
end
endmodule

```

Chapter 2

Verification, Co-Simulation & Waveform Analysis

2.1 IP Block 4: Digital Tachometer Verification

2.1.1 Digital Simulation (Icarus Verilog & GTKWave)

Before mixed-signal translation, the arithmetic accuracy and metastability protections of the pipeline were verified natively using a self-checking testbench in Icarus Verilog.

A 10 MHz (100ns period) system clock was synthesized alongside a simulated asynchronous sensor pulse arriving at exactly 500 μ s, representing a motor completing one rotation.



Figure 2.1: GTKWave Digital Simulation demonstrating the 16-bit hardware stopwatch capturing asynchronous motor pulses.

As evidenced by the digital waveform data (Figure 2.1):

1. **Synchronous Isolation:** The `sync` registers successfully isolate the high-speed counter from the unaligned `sensor_in` boundaries.
2. **Mathematical Accuracy:** At exactly 500 μ s, the motor pulse triggers the capture logic. The `period_count` output bus updates immediately to 1388₁₆. Converting this hexadecimal value to decimal yields exactly **5000**. Given the 100ns system clock period, $5000 \times 100\text{ns} = 500 \mu\text{s}$, proving the period tracking logic is mathematically flawless.

2.1.2 System Integration (eSim Mixed-Signal Scope)

To bridge this verified digital core into physical applications, the RTL was instantiated in the eSim NGVeri mixed-signal environment.

Schematic Wiring & ADC/DAC Boundaries

Analog `vpulse` sources were utilized to drive the system clock, reset, and motor sensor inputs. These analog signals were routed through a 3-bit `adc_bridge` to interface safely with

the digital tachometer block. On the output stage, the 16-bit `period_count` bus and the 1-bit `data_valid` strobe were routed through respective `dac_bridge` components to allow continuous analog visualization and extraction in Ngspice.

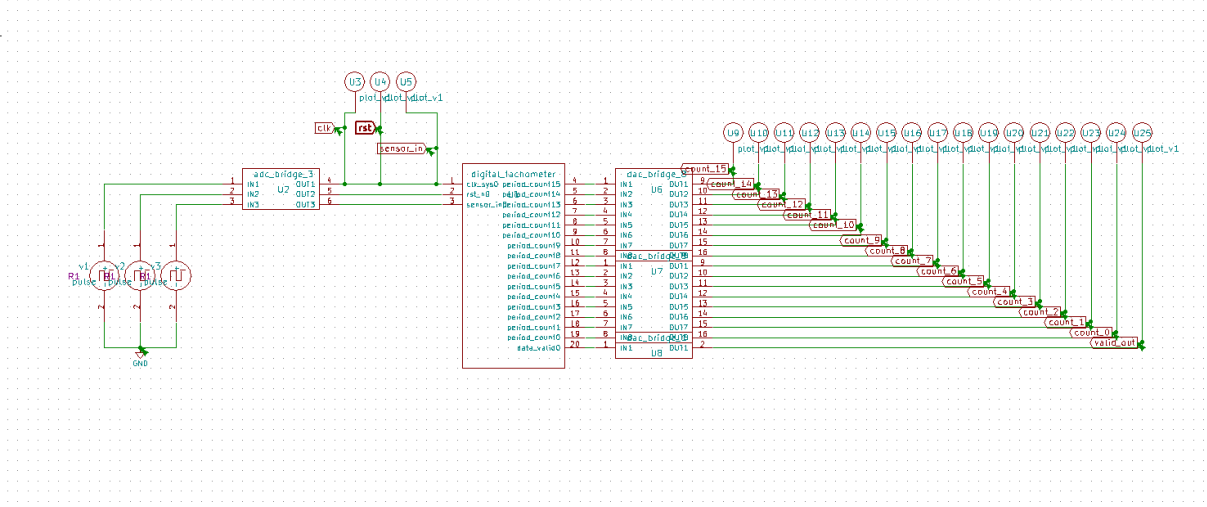


Figure 2.2: eSim Mixed-Signal Schematic of the Digital Tachometer with ADC and DAC conversion boundaries.

Advanced Ngspice Plot Customization

Visualizing a 16-bit digital bus in an analog simulator traditionally results in 16 overlapping 5V traces. To extract meaningful performance data, an advanced mathematical reconstruction was implemented directly within the Ngspice `.control` script. The 5V logic levels were normalized and multiplied by their binary weights to plot a direct decimal integer on the Y-axis.

```
* Ngspice Control Block for 16-bit Binary-to-Decimal Reconstruction
.control
run
print allv > plot_data_v.txt
print alli > plot_data_i.txt

set xbrushwidth=3
set color0=black
set color1=white

* Mathematically reconstruct the 16 individual 5V bits into a single integer
let count_val = (v(count_15)/5)*32768 + (v(count_14)/5)*16384 +
(v(count_13)/5)*8192 + (v(count_12)/5)*4096 + (v(count_11)/5)*2048 +
(v(count_10)/5)*1024 + (v(count_9)/5)*512 + (v(count_8)/5)*256 +
(v(count_7)/5)*128 + (v(count_6)/5)*64 + (v(count_5)/5)*32 +
(v(count_4)/5)*16 + (v(count_3)/5)*8 + (v(count_2)/5)*4 +
(v(count_1)/5)*2 + (v(count_0)/5)*1

* Plot the reconstructed integer and control signals
plot count_val
plot v(valid_out) (v(sensor_in)+6) (v(rst)+12)

.endc
```

Interpretation of Output Waveforms

The transient simulation verifies the precise mixed-signal boundary crossing of the high-speed data.

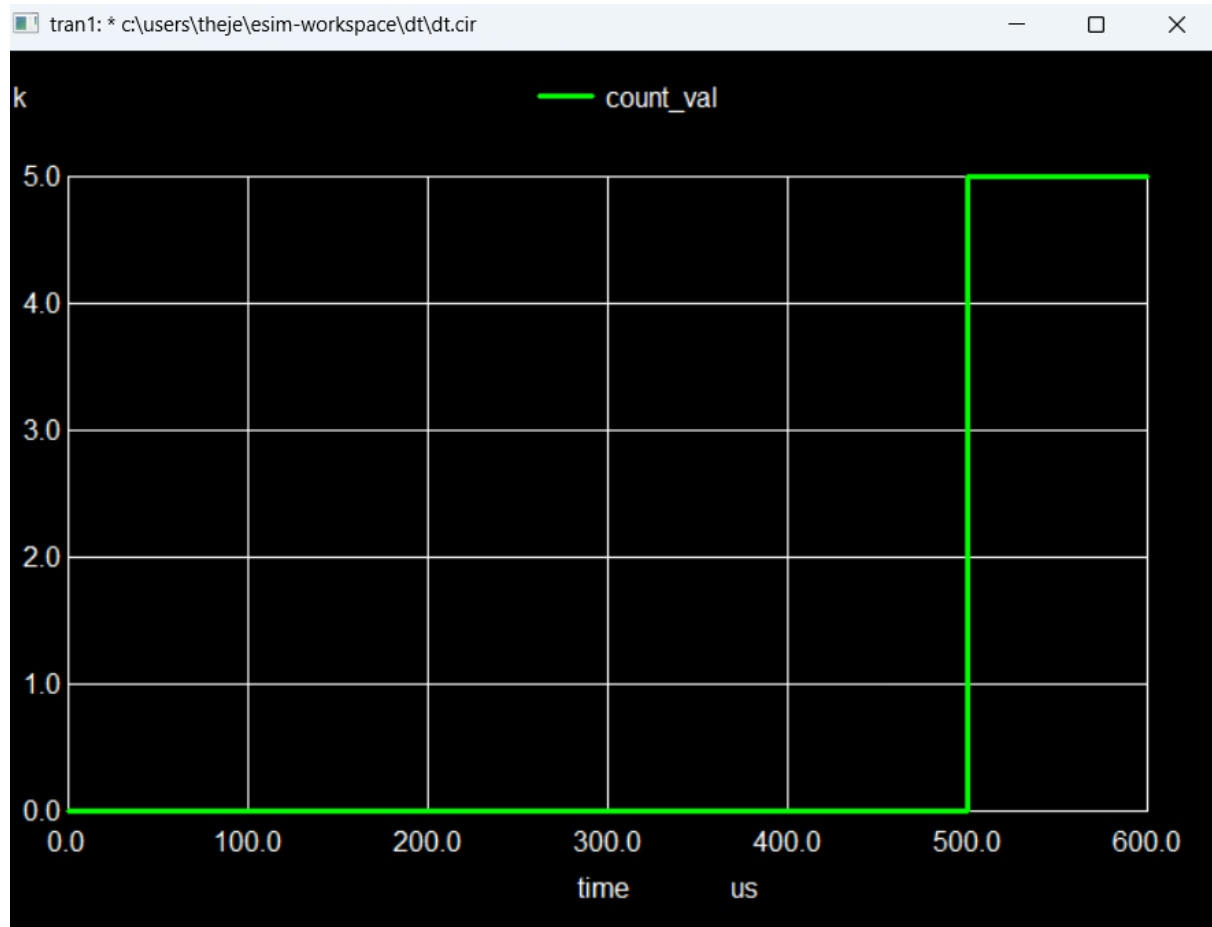


Figure 2.3: Ngspice Transient Simulation demonstrating the mathematically reconstructed decimal extraction of the 16-bit bus at the 500 μ s boundary.

As evidenced by the simulation data (Figures 2.3 and 2.4):

1. **Data Capture Validation:** At the 500 μ s mark, the `sensor_in` pulse triggers the synchronizer. Exactly 3 system clock cycles later, the `valid_out` DAC line cleanly spikes to 5V, confirming physical interrupt generation.
2. **Analog Bus Reconstruction:** The mathematically reconstructed `count_val` trace in the transient waveform plot steps instantaneously to 5.0k (5000), proving the 16 parallel DAC lines successfully transmitted the captured value without timing skew.
3. **Terminal Verification:** The Ngspice output terminal explicitly logs `period_count=5000`, independently confirming absolute parity between the digital Icarus simulation and the eSim analog domain.

```
ngspice -p C:\Users\Theje\Sim-Workspace\DT\DT.cir.out
clk_sys=1
rst_n=1
sensor_in=0
period_count=5000
data_valid=0

Inside foo after eval.....
clk_sys=1
rst_n=1
sensor_in=0
period_count=5000
data_valid=0
=====digital_tachometer : New Iteration=====
Instance : 0

Inside foo before eval.....
clk_sys=1
rst_n=1
sensor_in=0
period_count=5000
data_valid=0

Inside foo after eval.....
clk_sys=1
```

Figure 2.4: Ngspice Terminal Output explicitly logging the captured `period_count` variable during simulation execution.